



SCHNEIDER ELECTRIC: USING STATE-OF-THE-ART ACOUSTICAL IMAGING TECHNOLOGY FOR COMPRESSED AIR LEAK DETECTION AND REPAIR

OVERVIEW

Compressed air represents 1-15% of the total electricity consumption at Schneider Electric's U.S. sites, but leaks account for a significant source of wasted energy in a compressed air system and can be as much as 20-30% of the compressor's output. To accelerate the company's progress toward its energy goals, the Schneider team prioritized compressed air leak detection and adopted a new method using acoustic imaging technology. This approach uses audible and visual inputs and has the potential to significantly lower compressed air and process gas costs.



ORGANIZATIONAL TYPE

Industrial

BARRIER

Finding new cost-effective ways to detect compressed air leaks and improve overall system performance.

SOLUTION

Updating Schneider's standardized leak detection and repair process with the use of state-of-the-art acoustical imaging technology.

OUTCOME

The new leak detection process has identified more than 250 leaks across six sites, saving over \$60,000 in annual operating costs and reducing annual emissions by over 300 metric tons of CO₂.

POLICIES

Schneider documents compressed air system improvement processes in the company's Global Energy Best Practice Manual. The Manual has a dedicated section on compressed air system optimization.

PROCESS

Schneider had a challenging time in the past convincing stakeholders that compressed air leaks were a serious problem and that more advanced detection technology was worth the investment. Before implementing the acoustic imaging leak detector process, Schneider followed traditional approaches to identify compressed air leaks—applying soapy solutions to pipes and hoses and using human ears to detect leaks. Additionally, electronic ultrasonic audio instruments were used but these tools were not always able to locate the precise location of leaks. Schneider worked with a third party to evaluate state-of-the-art acoustical imaging technology. These tools use a video camera to provide a live image, and an array of 64 different directional microphones to create a heat map of ultrasonic emissions on top of the visual display. Using audible and visual inputs, the acoustical imaging technology enables maintenance teams to locate compressed air, gas, and vacuum leaks quickly and accurately.

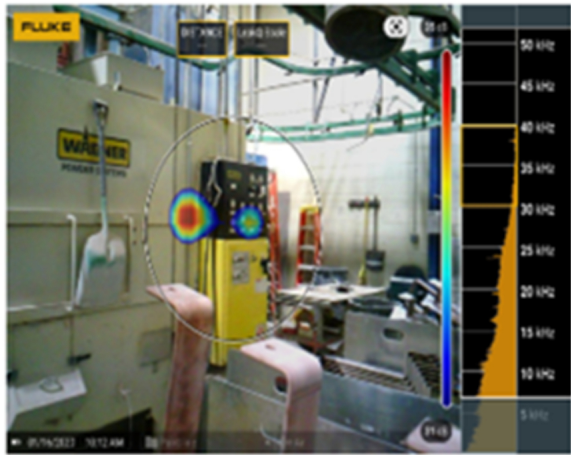
Once the tools were rolled out with training and assistance from their Integrated Facilities Management (IFM) partner, a standardized process was developed to instruct site-level personnel to perform the facility's compressed air assessment and track data monthly. Site facilities personnel use a reporting tool to quantify leaks and savings by categorizing leaks into the appropriate areas of responsibility, either Production or Facilities. Then, the tool is shipped to the next scheduled site and the process is replicated.

Leak Survey and Fluke Meter schedule

Month/Yr	Meter	Date received	Priority	Property Name	Address 1	City	St	Country	Postal Code	Facility Manager	Complete	Ship to	Date Shipped	Notes
Jan 1 2022	Meter 1	1/1/22	5	Cedar Rapids	3700 6th St SW	Cedar Rapids	IA	United States	52404-5403	Dan Smith	Complete	Oxford	2/21/22	2 units 2019 and 20 for production
Feb 1 2022	Meter 1	2/2/22	5	Oxford	5735 College Corner Rd	Oxford	MO	United States	40506	Tom Messon	In progress	El Paso 1		5 units Total
Mar 1 2022	Meter 1		5	El Paso 1	1601 Northwestern Dr	El Paso	TX	United States	79912-8024	Aaron Hudak		El Paso 2		Do not have probe
Apr 1 2022	Meter 1		5	El Paso 2	7801 Northern Pass Dr	El Paso	TX	United States	79911	Aaron Hudak		El Paso 3		1 large unit, needs a Backup unit
Apr 1 2022	Meter 1		5	El Paso 3	1701 Northwestern Dr	El Paso	TX	United States	79912-1119	Aaron Hudak		Athens		3 units
May 1 2022	Meter 2		5	Athens	1651 Enterprise St	Athens	TX	United States	75751-8839	Lisa Cobb		Costa Mesa		
June 1 2022	Meter 1		5	Costa Mesa	1660 Scenic Ave	Costa Mesa	CA	United States	92626-1420	Aaron Hudak		Chino		1 unit NOT a large area
July 1 2022	Meter 1		5	Chino	14725-14875 Monte Vista Ave	Chino	CA	United States	91710-5732	Aaron Hudak		Lake Forest		
Aug 1 2022	Meter 1		2	Lake Forest 3	26521 Rancho Pkwy S	Lake Forest	CA	United States	92630-8329	Aaron Hudak		Portland		2 units
Sept 1 2022	Meter 1		5	Portland/Tualatin	12343 SW Leviston Dr	Tualatin	OR	United States	97062	Lisa Cobb		Victoria CAD		Do not have probe
Oct 1 2022	Meter 1		3	Victoria CAD - Keating	2195 Keating Cross Rd	Seasichton	BC	Canada	V8M2A5	Stefan Mathew		Crescent Park		Do not have probe
Nov 1 2022	Meter 1		5	West Chester OH - Crescent Park	9870 Crescent Park Dr	West Chester	OH	United States	43089-3800	TBO/Jimmy Levelace		Windloch Rd		4 big in power house. 3 small in the building order units
Dec 1 2022	Meter 1		5	Windloch Road	9701 Windloch Road	West Chester	OH	United States	43089-3800	TBO/Jimmy Levelace		Fairfield		2 new units 2018 and 2019
Jan 1 2023	Meter 1		2	Fairfield	5855 Union Centre Blvd	Fairfield	OH	United States	43024-2343	TBO/Jimmy Levelace		Huntington		
Feb 1 2023	Meter 1		5	Huntington	6 Commercial Rd	Huntington	IN	United States	46750-8805	TBO/Jimmy Levelace		Cedar Rapids		
Jan 1 2022	Meter 2	1/1/22	5	Seneca 1	1990 Sandifer Blvd	Seneca	SC	United States	29678	Joseph D'Allesman	Complete	Seneca 2		5- units with backup
Feb 1 2022	Meter 2	2/1/22	2	Seneca 2	2321 Blue Ridge Blvd	Seneca	SC	United States	29672-6939	Joseph D'Allesman	Complete	Smyrna	3/1/22	None probe
Mar 1 2022	Meter 2	3/4/22	1	Smyrna	530 Wexley Ln	Smyrna	TN	United States	37587-2024	James Hobden		Huntley		None probe
Mar 1 2022	Meter 2		1	Huntley	530 Huntley Industrial Dr	Smyrna	TN	United States	37587	James Hobden		Middlebroom		
Apr 1 2022	Meter 2		3	Middlebroom	203 Pulling Mill Rd	Middlebroom	PA	United States	17057-2921	Jennifer Reisinger		Mechanicsburg		3 units
May 1 2022	Meter 2		3	Mechanicsburg	203 Cummins/Plum Plany	Mechanicsburg	PA	United States	17055-5664	Jennifer Reisinger		Brossard 3		3 only 3 are working
June 1 2022	Meter 2		3	Brossard 3	4150 Place de Jaws	Brossard	QC	Canada	J4V 0C4	Ron Fisher		Brossard 1		
July 1 2022	Meter 2		1	Brossard 1	4200 Place de Jaws	Brossard	QC	Canada	J4V 0C4	Ron Fisher		Mississauga		
Aug 1 2022	Meter 2		1	Mississauga - McLaughlin 1	5985 McLaughlin Rd	Mississauga	ON	Canada	L5R 1R6	Ron Fisher		Wellcome		1 units with 100 Back-up. 500W
Sept 1 2022	Meter 2		5	Wellcome	325 Wellcome Center Blvd	Lexington	NC	United States	27295-9038	David Nasent		Selisbury		4- units with backup
Oct 1 2022	Meter 2		2	Selisbury	105 Summit Park Dr	Selisbury	NC	United States	28146-6325	David Nasent		Greensboro		
Nov 1 2022	Meter 2		5	Greensboro	215 Burgess Rd	Greensboro	NC	United States	27409-4778	David Nasent		Lincoln		
Dec 1 2022	Meter 2	Done for 2022	5	Lincoln	1717 Center Park Rd	Lincoln	NE	United States	68512-1231	Russ Hobson	Yes for 2022	Columbia, MO		1 unit
Jan 1 2023	Meter 1	Done for 2022	5	Columbia, MO	4800 Paris Rd	Columbia	MO	United States	65202-9396	Russ Hobson	Yes for 2022	Ofallon		
Feb 1 2023	Meter 1	Done for 2022	5	Ofallon	807 Corporate Centre Dr	Ofallon	MO	United States	63368-8885	Dan Smith	Yes for 2022	Columbia, SC		Does NOT use on property except to blow out computers as needed
Mar 1 2023	Meter 2	Done for 2022	5	Columbia, SC	8821 Gamers Ferry Rd	Columbia	SC	United States	29061	Joseph D'Allesman	Yes for 2022	Seneca 1		
Apr 1 2023	Meter 2	Done for 2022	5	Lexington	1401 Mercer Rd	Lexington	KY	United States	40511-5325	Marc Hill	Yes for 2022	Lexington has their own meter		2 units, 5 HP in systems

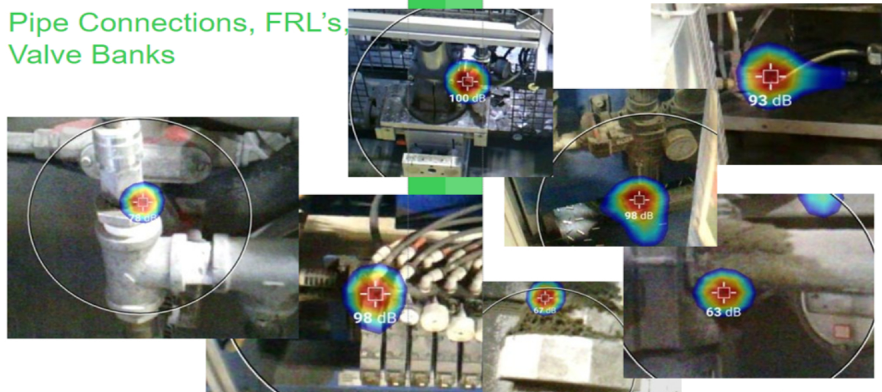
Leak Details

File	Paint line _0004.as2
Leak Number	4
Type	LeakQ Rate Capture
Folder Name	Paint line
Date and Time	1/16/2020 4:12:32 AM
Distance	11.8 ft
LeakQ Rate	5.8
Measured dB Level	56.6
Estimated Leak Rate	0.5 CFM
Estimated Costs	256 USD/year





Pipe Connections, FRL's,
Valve Banks



MEASURING SUCCESS

A report generation tool quantifies leaks and savings using the acoustical imager, which calculates potential energy savings and CO₂ emissions reductions. The standardized process instructs site-level personnel to perform the assessment and save the data within one month, then have the tool shipped to the next site on the schedule.

OUTCOMES

To date, 40 Schneider staff have been trained to use the acoustic imager tools for compressed air leak detection. Over 250 leaks across six facilities have been identified, leading to over \$60,000 in annual cost savings. By repairing the detected leaks, Schneider Electric will reduce its annual emissions by over 300 metric tons of CO₂. The success of this approach improved awareness of the losses from compressed air leaks and has increased management buy-in. The use of acoustic technology is not limited to compressed air and can detect nitrogen, bulk gas, steam, and vacuum leakage, as well as issues associated with pump cavitation, noisy bearings, and electrical discharge. The use of Schneider's improved leak detection process will be expanded to detect issues from other equipment that generate acoustic signals going forward.